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TANK BREATHER CAP WITH INTEGRATED FILTER, SPLASH PROTECTION, AND NIPPLE FOR BREATHER HOSE

Abstract

A hydraulic fluid tank assembly for use with a suspension system for a vehicle is provided. The hydraulic fluid tank assembly includes a tank having a chamber configured to contain a volume of hydraulic fluid and a volume of gas, and a breather cap assembly. The breather cap assembly includes a breather cap having a passageway and a recess, a filter having one or more openings configured to permit gas to pass therethrough, and a splash guard coupled to the breather cap, the splash guard retaining the filter in the recess of the breather cap, and the splash guard having a distal opening. The passageway of the breather cap, the openings of the filter, and the distal opening of the splash guard are fluidly connected and configured to permit gas to pass therethrough to and from the chamber of the tank and the atmosphere.

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Background/Summary

FIELD

[0001] The present disclosure relates generally to suspension systems for motor vehicles and more particularly to hydraulic fluid tank and breather cap assemblies used with various suspension systems.

BACKGROUND

[0002] The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

[0003] Suspension systems improve the ride of a vehicle by absorbing bumps and vibrations that would otherwise unsettle the vehicle body. Suspension systems also improve safety and control by improving contact between the ground and the tires of the vehicle. One drawback of suspension systems is that basic spring/damper arrangements will allow the vehicle to roll/lean right or left during cornering (e.g., in turns), pitch forward under deceleration (e.g., under braking), and pitch back under acceleration. The lateral acceleration the vehicle experiences in turns causes a roll moment where the vehicle will lean/squat to the right when turning left and to the left when turning right. The fore and aft acceleration the vehicle experiences under acceleration and braking causes a pitch moment where the vehicle will lean forward loading the front axle during braking and aft, loading the rear axle, under acceleration. These roll and pitch moments decrease grip, cornering performance, and braking performance and can also be uncomfortable to the driver and passengers.

[0004] Many vehicles are equipped with stabilizer bars/anti-roll bars, which are mechanical systems that help counteract the roll moments experienced during driving. For example, anti-roll bars are typically mechanical linkages that extend laterally across the width of the vehicle between the right and left dampers. When one of the dampers extends, the anti-roll bar applies a force to the opposite damper that counteracts the roll moment of the vehicle and helps to correct the roll angle to provide flatter cornering. However, there are several drawbacks associated with these mechanical systems. First, there are often packaging constraints associated with mechanical systems because a stabilizer bar/anti-roll bar requires a relatively straight, unobstructed path across the vehicle between the dampers. Second, stabilizer bars/anti-roll bars are reactive and work when the suspension starts moving (i.e. leaning). Such mechanical systems cannot be easily switched off or cancelled out when roll stiffness is not needed. Some vehicles do have stabilizer bar/anti-roll bar disconnects that may be manually or electronically actuated, but the complexity and costs associated with these systems make them ill-suited for most vehicle applications. Packaging constraints also limit the ability to provide mechanical systems that effectively limit fore and aft pitch.

[0005] In an effort to augment or replace traditional mechanical stabilizer bars/anti-roll bars, anti-roll suspension systems are being developed that include one or more dampers that be independently actuated via one or more fluidly-connected pumps, wherein the pump(s) can produce a pressure change in the damper(s). In some suspension systems, there is one pump per damper, whereas in other suspension systems, there is one pump per two or more dampers. Additionally, other anti-roll suspension systems are being developed that hydraulically connect two, or more dampers in a hydraulic circuit where the extension of one damper produces a pressure change in the other damper(s) in the hydraulic circuit that makes it more difficult to compress the other damper(s) in the hydraulic circuit. The pressure changes in the damper(s) increases the roll and pitch stiffness of the suspension system of the vehicle. As stated herein these suspension systems use pumps to generate the pressure changes in the dampers and the active forces exerted by the dampers. The flow or the speed of the pump is controlled to generate a requested pressure which is translated to a force by the damper.

[0006] The pumps rely on a source of hydraulic fluid, which is contained in a hydraulic fluid tank onboard the vehicle. The hydraulic fluid volume can in some instances be large, and therefore, the hydraulic fluid tank needs to be ventilated to atmospheric pressure. Otherwise, the tank would become a pressure vessel with high pressure when an amount of hydraulic fluid is pumped into or out of the tank. Accordingly, such hydraulic fluid tanks are provided with a breather that allows the interior of the tank to be open to the environment. However, the breather needs to have a filter to protect the hydraulic fluid from contamination. If contamination gets into the hydraulic fluid inside the tank, there is a risk that the contamination will reduce the performance of the suspension system.

[0007] Accordingly, there is a need for a suspension system that includes a breather that better protects the hydraulic fluid in the tank from contamination.

SUMMARY

[0008] This section provides a general summary of the disclosure, and is not a comprehensive disclosure of its full scope or all of its features.

[0009] In accordance with one aspect of the subject disclosure, a hydraulic fluid tank assembly for a suspension system for a vehicle is provided. The hydraulic fluid tank assembly includes a tank and a breather cap assembly. The breather cap assembly has a low restriction so that over/under pressure is not created in the tank when hydraulic fluid is pumped into or out of the tank. The breather cap assembly includes a breather cap, a filter, and a splash guard. The filter protects the hydraulic fluid in the tank from contamination entering the tank through the breather assembly. The breather cap assembly may further include an attachment point for a breather hose, wherein the outlet/inlet of the breather hose is positioned on the vehicle such that the inlet/outlet of the breather hose is always above the maximum water wading depth of the vehicle. When the vehicle is driving in extreme conditions, for example up and/or down steep inclines, off-road, or on racetracks, the hydraulic fluid sloshes around in the tank. The splash guard is located between the filter and the hydraulic fluid in the tank. The splash guard reduces or prevents the potential for hydraulic fluid to splash and contact the filter. The splash guard also retains the filter in the breather cap.

[0010] In accordance with another aspect of the subject disclosure, a hydraulic fluid tank assembly for use with suspension system of a vehicle is provided. The hydraulic fluid tank assembly includes a tank and a breather cap assembly releasably coupled to the tank. The tank has a chamber configured to contain a volume of hydraulic fluid and a volume of gas. The breather cap assembly includes a breather cap, the breather cap having a passageway and a recess, a filter, the filter having one or more openings configured to permit gas to pass therethrough, and a splash guard coupled to the breather cap, the filter disposed between the splash guard and the breather cap, the splash guard retaining the filter in the recess of the breather cap, and the splash guard having a distal opening. The passageway of the breather cap, the openings of the filter, and the distal opening of the splash guard are fluidly connected and configured to permit gas to pass therethrough to and from the chamber of the tank and the atmosphere.

[0011] In accordance with another aspect of the subject disclosure, a hydraulic fluid tank assembly for use with suspension system of a vehicle is provided. The hydraulic fluid tank assembly includes a tank and a breather cap assembly releasably coupled to the tank. The tank has a chamber configured to contain a volume of hydraulic fluid and a volume of gas. The breather cap assembly includes a breather cap, a filter, and a splash guard. The breather cap includes a base having a first side and a second side opposite the first side, an outer cylindrical skirt extending a first distance from the second side of the base away from the first side, an inner side of the outer cylindrical skirt having a retaining feature adapted to releasably couple the breather cap to the tank, an inner cylindrical skirt extending a second distance from the second side of the base away from the first side, the second distance less than the first distance, and a passageway extending through the base. The filter has one or more openings configured to permit gas to pass therethrough. The splash guard has a distal opening and is coupled to the inner cylindrical skirt of the breather cap, with the

filter disposed between the splash guard and the breather cap. The passageway of the breather cap, the openings of the filter, and the distal opening of the splash guard fluidly connected and configured to permit gas to pass therethrough to and from the chamber of the tank and the atmosphere.

[0012] In accordance with another aspect of the subject disclosure, a hydraulic fluid tank assembly for use with a suspension system for a vehicle is provided. The hydraulic fluid tank assembly includes a tank, the tank having a chamber configured to contain a volume of hydraulic fluid and a volume of gas, and a breather cap assembly releasably coupled to the tank. The breather cap assembly includes a breather cap, a filter, and a splash guard. The breather cap includes a base having a first side and a second side opposite the first side, a cylindrical skirt extending from the second side of the base and away from the first side, the cylindrical skirt having a first outer side, a second inner side, and a shoulder on the second inner side of the cylindrical skirt, the shoulder located a distance from the second side of the base, one or more grooves on the second inner side of the cylindrical skirt, the one or more grooves proximate to the shoulder, and a passageway extending through the base from the first side of the base to the second side of the base. The filter includes a filter edge finishing and a filter element, the filter edge finishing in contact with the shoulder of the breather cap, the filter configured to permit gas to pass therethrough. The splash guard is coupled to the breather cap, with the filter disposed between the splash guard and the breather cap. The splash guard includes a circular base having a first side and a second side opposite the first side, one or more tabs extending radially from circular base, the one or more tabs located in the one or more grooves of the breather cap, a guard portion extending from the second side of the base away from the first side, the guard portion extending toward the chamber of the tank, and a distal opening extending through the guard portion.

[0013] Further areas of applicability and advantages will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations, and are not intended to limit the scope of the present disclosure.

[0015] FIG. 1 is an illustration of an exemplary vehicle equipped with a suspension system in accordance with the teachings of the present disclosure;

[0016] FIG. 2 is a schematic diagram illustrating a portion of the suspension system in accordance with the teachings of the present disclosure;

[0017] FIG. 3 is a perspective view of a portion of an exemplary hydraulic fluid tank assembly having a hydraulic fluid tank and a breather cap assembly in accordance with the teachings of the present disclosure;

[0018] FIG. 4 is a section view of the hydraulic fluid tank assembly illustrated in FIG. 3;

[0019] FIG. 5 is an exploded top perspective view of an exemplary breather cap assembly having a breather cap, a filter, and a splash guard in accordance with the teachings of the present disclosure;

[0020] FIG. 6 is an exploded bottom perspective view of the exemplary breather cap assembly shown in FIG. 5;

[0021] FIG. 7 is an exploded cross-sectional view of the exemplary breather cap assembly shown in FIG. 5;

[0022] FIG. 8 is a perspective section view of the exemplary breather cap shown in FIG. 5;

[0023] FIG. 9 is a bottom view of the exemplary splash guard shown in FIG. 5;

[0024] FIG. **10** is a perspective section view of the exemplary breather cap assembly shown in FIG. **5**;

[0025] FIG. **11** is a bottom perspective section view of the exemplary breather cap assembly shown in FIG. **5**;

[0026] FIG. **12** is a cross-sectional view of the exemplary breather cap assembly shown in FIG. **5** having an exemplary optional dipstick;

[0027] FIG. **13** is a section view of another exemplary hydraulic fluid tank assembly having a hydraulic fluid tank and a breather cap assembly in accordance with the teachings of the present disclosure;

[0028] FIG. **14** is an exploded top perspective view of the exemplary breather cap assembly shown in FIG. **13** having a breather cap, a filter, and a splash guard in accordance with the teachings of the present disclosure;

[0029] FIG. **15** is an exploded bottom perspective partial section view of the exemplary breather cap assembly shown in FIG. **14**;

[0030] FIG. **16** is an exploded cross-sectional view of the exemplary breather cap assembly shown in FIG. **14**;

[0031] FIG. **17** is a cross-sectional view of the exemplary breather cap assembly shown in FIG. **14**; and

[0032] FIG. **18** is a top view of the exemplary breather cap assembly shown in FIG. **14**.

[0033] Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

[0034] The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses.

[0035] Example embodiments are provided so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.

[0036] The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

[0037] When an element or layer is referred to as being “on,” “engaged to,” “connected to,” or “coupled to” another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the

associated listed items. As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

[0038] Although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.

[0039] Spatially relative terms, such as “inner,” “outer,” “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0040] With reference to FIG. 1, an exemplary vehicle **10** is illustrated that has been equipped with a suspension system **12** of the present disclosure. The vehicle **10** in FIG. 1 has been illustrated as an automobile; however, it should be appreciated that the suspension system **12** described herein is not limited to automobiles and may be used in other types of vehicles. In the illustrated example, the vehicle **10** has four wheels **14**; however, it should be appreciated that the vehicle **10** may have fewer or greater than four wheels. Similarly, the suspension system **12** of the vehicle **10** includes a plurality of dampers **16a-16d**, with one damper **16a-16d** per wheel **14**, including a front left damper **16a**, a front right damper **16b**, a back left damper **16c**, and a back right damper **16d**. While it should be appreciated that the suspension system **12** described herein may include a different number of dampers than those shown in the drawings, in most automotive applications, four dampers are used at each corner of a vehicle to control vertical movements of the front and rear wheels of the vehicle. Thus, the front left damper **16a** controls (e.g., dampens) up and down (i.e., vertical) movements of the front left wheel **14** of the vehicle **10**, the front right damper **16b** controls (e.g., dampens) up and down (i.e., vertical) movements of the front right wheel **14** of the vehicle, the back left damper **16c** controls (e.g., dampens) up and down (i.e., vertical) movements of the back left wheel of the vehicle **10**, and the back right damper **16d** controls (e.g., dampens) up and down (i.e., vertical) movements of the back right wheel of the vehicle **10**.

[0041] The suspension system **12** of the vehicle **10** also includes a pump assembly **18** and a hydraulic fluid tank assembly **20**. The pump assembly **18**, is fluidly connected to the hydraulic fluid tank assembly **20**, the front left damper **16a**, the front right damper **16b**, the back left damper **16c**, and the back right damper **16d**. Although the suspension system **12** in FIG. 1 is shown having a single pump assembly **18** and a single hydraulic fluid tank assembly **20**, it should be appreciated that in some embodiments, the suspension system **12** may include two pump assemblies **18** and two hydraulic fluid tank assemblies **20**, comprising a front pump assembly **18** fluidly connected to a front hydraulic fluid tank assembly **20**, the front left damper **16a**, and the front right damper **16b** and a back pump assembly **18** fluidly connected to a back hydraulic fluid tank assembly **20**, the back left damper **16c**, and the back right damper **16d**. In other embodiments, the suspension system **12** may include two pump assemblies, comprising a first pump assembly **18** fluidly connected to a first hydraulic fluid tank assembly **20**, the front left damper **16a**, and the back right damper **16d** and a second pump assembly **18** fluidly connected to a second hydraulic fluid tank assembly **20**, the

front right damper **16b**, and the back left damper **16c**. In yet other embodiments, the suspension system **12** may include four pump assemblies **18** and four hydraulic fluid tank assemblies **20**, with one pump assembly **18** and hydraulic fluid tank assembly **20** per damper **16a-16d**.

[0042] The fluid pressure in the dampers **16a**, **16b**, **16c**, **16d** can be dynamically adjusted by the pump assembly **18** to dynamically adjust the roll stiffness of the vehicle **10** and can be used to either augment or completely replace mechanical stabilizer bars/anti-roll bars. Such mechanical systems require relatively straight, unobstructed runs between each of the front dampers **16a**, **16b** and each of the back dampers **16c**, **16d**. Accordingly, the suspension system **12** disclosed herein offers packaging benefits because the dampers **16a**, **16b**, **16c**, **16d** do not need to be mechanically connected.

[0043] In the illustrated embodiment, the dampers **16a**, **16b**, **16c**, **16d** are identical; however, other configurations are possible where the front dampers **16a**, **16b** differ from the rear dampers **16c**, **16d**.

[0044] The suspension system **12** also includes a suspension control unit (SCU) **22** that includes one or more processors or controllers configured to execute computer programs to control the suspension system **12** by implementing the control methods described below and memory that is programmed with the aforementioned computer programs and control methods. The suspension control unit (SCU) **22** is configured, for example, to control the pump assembly **18** to pump hydraulic fluid into and/or out of the dampers **16a**, **16b**, **16c**, **16d** and hydraulic fluid tank assembly **20**.

[0045] With reference to FIG. 2, additional details of the suspension system **12** are described with respect to a single damper **16**, a single pump assembly **18**, and a single hydraulic fluid tank assembly **20**; however, it should be understood that the description with respect to the single damper **16** applies to the dampers **16a**, **16b**, **16c**, **16d**. The damper **16** is fluidly connected to the pump assembly **18** and the hydraulic fluid tank assembly **20**. The damper **16** includes a damper housing **24**, a piston rod **26**, and a piston **28** that is mounted on the piston rod **26**. The piston **28** is arranged in sliding engagement with the inside of the damper housing **24** such that the piston **28** divides the damper housing **24** into a first working chamber **30** and a second working chamber **32**.

[0046] The pump assembly **18a** includes a pump **34** and a pump motor **36**. The pump **34** has a first pump port **38** and a second pump port **40**. The pump **34** is a bi-directional pump driven by the pump motor **36**.

[0047] The suspension system **12** further includes a plurality of hydraulic lines **42**, **44**, **46**, **48**, **50**, and **52** fluidly connecting the pump **34**, the hydraulic fluid tank assembly **20**, the first working chamber **30**, and the second working chamber **32**. For example, the first pump port **38** is fluidly connected to the first working chamber **30** of the damper **16** by the first hydraulic line **42** and the third hydraulic line **46**, and the second pump port **40** is fluidly connected to the second working chamber **32** of the damper **16** by the second hydraulic line **44** and the fourth hydraulic line **48**. The first hydraulic line **42** and the third hydraulic line **46** are fluidly connected to one another. The second hydraulic line **44** and the fourth hydraulic line **48** are fluidly connected to one another. Although the first hydraulic line **42** and the third hydraulic line **46** are referenced as two hydraulic lines, it will be understood that in some embodiments, the first hydraulic line **42** and the third hydraulic line **46** are a single hydraulic line. Additionally, although the second hydraulic line **44** and the fourth hydraulic line **48** are referenced as two hydraulic lines, it will be understood that in some embodiments, the second hydraulic line **44** and the fourth hydraulic line **48** are a single hydraulic line.

[0048] Additionally, the first working chamber **30**, the first pump port **38**, and the hydraulic fluid tank assembly **20** are fluidly connected to one another by the first hydraulic line **42**, the third hydraulic line **46**, and the fifth hydraulic line **50**. The second working chamber **32**, the second pump port **40**, and the hydraulic fluid tank assembly **20** are fluidly connected to one another by the second hydraulic line **44**, the fourth hydraulic line **48**, and the sixth hydraulic line **52**. It should be

appreciated that the hydraulic lines **42**, **44**, **46**, **48**, **50**, and **52** may be constructed from rigid line, flexible tubing, hoses, integrated into the pump assembly **18** or dampers **16**, or the like.

[0049] The suspension system **12** further includes a first valve **54** and a second valve **56**. The first valve **54** is fluidly connected with the pump **34**, the hydraulic fluid tank assembly **20**, and the first working chamber **30**. The second valve **56** is fluidly connected with the pump **34**, the hydraulic fluid tank assembly **20**, and the second working chamber **32**. As shown in FIG. 2, the first valve **54** is in line with the fifth hydraulic line **50**, and when open, permits hydraulic fluid to flow from the hydraulic fluid tank assembly **20** through the fifth hydraulic line **50**. The second valve **56** is in line with the sixth hydraulic line **52**, and when open, permits hydraulic fluid to flow from the hydraulic fluid tank assembly **20** through the sixth hydraulic line **52**.

[0050] In some embodiments, the first valve **54** and the second valve **56** are substantially similar to each other. In some embodiments, each of the first valve **54** and the second valve **56** are flow control valves. The first valve **54** and the second valve **56** may be electro-mechanical valves that are semi-active valves with a combination of passive spring-disk elements and a solenoid. An electrical current may be supplied to the solenoids of the first valve **54** and/or the second valve **56** to open the first valve **54** and/or the second valve **56** to change the damping characteristics of the damper **16**. Although the first valve **54** and the second valve **56** are described as solenoid operated, they may be operated by other electrical structures or using energy other than electrical energy. In some embodiments, the first valve **54** and the second valve **56** may be configured as hydraulic passive valves or semi-active hydraulic valves. As such, the first valve **54** and the second valve **56** need not be in receipt of an electrical signal if configured as passive valves.

[0051] Having described the components of an exemplary embodiment of the damper **16**, operation of the damper **16** is described. It should be appreciated that because the pump **34** is a bi-directional pump, the pump port that acts as an intake on the pump **34** and the pump port and that acts as an outlet on the pump **34** may be switched when the rotational direction of the pump motor **36** is reversed. For example, when the pump motor **36** is driving in a first direction and the first valve **54** is open, fluid is drawn from the hydraulic fluid tank assembly **20** through the fifth hydraulic line **50** and the third hydraulic line **46** to the first pump port **38**. The first pump port **38** acts as a pump intake during this portion of operation. Pressurized fluid exits the second pump port **40** acting as a pump outlet. In a different operational mode, when the pump motor **36** is driving in a second direction (opposite the first direction) and the second valve **56** is open, fluid is drawn from the hydraulic fluid tank assembly **20** through the sixth hydraulic line **52** and the fourth hydraulic line **48** to the second pump port **40**. The second pump port **40** acts as a pump intake during this portion of operation. Pressurized fluid exits the first pump port **38** acting as a pump outlet.

[0052] As an additional example, when the pump motor **36** is driving in a first direction and the first valve **54** and the second valve **56** are closed, fluid is drawn from the first working chamber **30** through the first hydraulic line **42** and the third hydraulic line **46** to the first pump port **38**. The first pump port **38** acts as a pump intake during this portion of operation. Pressurized fluid exits the second pump port **40** acting as a pump outlet and is directed to the second working chamber **32** through the fourth hydraulic line **48** and the second hydraulic line **44**. In a different operational mode, when the pump motor **36** is driving in a second direction (opposite the first direction) and the first valve **54** and the second valve **56** are closed, fluid is drawn from the second working chamber **32** through the second hydraulic line **44** and the fourth hydraulic line **48** to the second pump port **40**. The second pump port **40** acts as a pump intake during this portion of operation. Pressurized fluid exits the first pump port **38** acting as a pump outlet and is directed to the first working chamber **30** through the third hydraulic line **46** and the first hydraulic line **42**.

[0053] It will be understood that the components and fluid connections shown in FIG. 2 are merely exemplary and that additional or alternative components and/or fluid connections may be used without departing from the scope of the disclosure.

[0054] With reference to FIGS. 3 and 4, an example of the hydraulic fluid tank assembly **20** is

illustrated. The hydraulic fluid tank assembly **20** includes a hydraulic fluid tank **58**, a breather cap assembly **60**, and a gasket **62**. The hydraulic fluid tank **58** has an interior chamber **64** configured to contain a volume of hydraulic fluid and a volume of gas (e.g., air). The breather cap assembly **60** is releasably coupled to the hydraulic fluid tank **58** and may be removed from and replaced onto the hydraulic fluid tank **58** to enable maintenance of the breather cap assembly **60**, such as for example replacement of one or more components of the breather cap assembly **60**. In some embodiments, the breather cap assembly **60** may be removed to enable filling of the hydraulic fluid tank **58** with hydraulic fluid. The gasket **62** located between the hydraulic fluid tank **58** and the breather cap assembly **60** to prevent leaking of hydraulic fluid and/or gas out of the hydraulic fluid tank **58** between the interface of the hydraulic fluid tank **58** and the breather cap assembly **60**.

Breather Cap Assembly

[0055] With reference to FIGS. **5**, **6**, **7**, **10**, **11**, and **12**, an embodiment of the exemplary breather cap assembly **60** is shown. The breather cap assembly **60** includes a breather cap **66**, a filter **68**, and a splash guard **70**. The filter **68** is sandwiched between the breather cap **66** and the splash guard **70**.

Breather Cap

[0056] With additional reference to FIGS. **5**, **6**, **7**, and **8**, the breather cap **66** includes a base **72**, a skirt **74** extending from the base **72**, and a passageway **76** extending through the base **72**. The base **72** has a first side **78**, a second side **80** opposite the first side **78**, and a thickness **82** between the first side **78** and the second side **80**. In some embodiments, the base **72** is cylindrical and has a first outer diameter **84**. Although the base **72** is shown as cylindrical, the base **72** may have other shapes without departing from the scope of the disclosure.

[0057] The skirt **74** extends a first distance **86** from the second side **80** of the base **72** and away from the first side **78** of the base **72**. The skirt **74** terminates in a distal end **88**. The exemplary skirt **74** is cylindrical and includes a first outer side **90** and a second inner side **92**. The first outer side **90** of the skirt **74** has a second outer diameter **94**, wherein the second outer diameter **94** of the skirt **74** is less than the first outer diameter **84** of the base **72**. The second side **80** of the base **72** forms a first shoulder **96**. The gasket **62** is sized to fit over the second outer diameter **94** of the skirt **74** and is configured to be sandwiched between the first shoulder **96** of the base **72** and the hydraulic fluid tank **58** when the breather cap assembly **60** is secured to the hydraulic fluid tank **58**.

[0058] At the distal end **88** of the skirt **74** is a distal opening **97** to an open chamber **98** formed by the skirt **74**. The open chamber **98** may also be referred to as a recess or a pocket. Additionally, at the distal end **88** of the skirt **74** is an annular end surface **100** extending between the first outer side **90** and the second inner side **92**. The breather cap **66** further includes a second shoulder **102** on the second inner side **92** of the skirt **74**. The second shoulder **102** is located a second distance **104** from the second side **80** of the base **72**, where the second distance **104** is less than the first distance **86**. Thus, the second shoulder **102** is located between the second side **80** of the base **72** and the distal end **88** of the skirt **74**. The second shoulder **102** is also located a third distance **106** inward from the distal end **88** of the skirt **74** toward the second surface **80** of the base **72**. The third distance **106** may be equal to the difference between the first distance **86** and the second distance **104**. The breather cap **66** further includes a nipple **108** extending from the first side **78** of the base **72** away from the second side **80** of the base **72**. The passageway **76** extends through the nipple **108** and the base **72**. The nipple **108** is configured to be fluidly coupled with a breather hose (not shown), wherein the breather hose terminates in an opening that is above the maximum water wading depth of the vehicle **10**. This reduces or prevents the possibility of water entering the hydraulic fluid tank **58** if the vehicle **10** enters water.

[0059] The breather cap **66** includes one or more fastening features configured to releasably couple the breather cap **66** to the hydraulic fluid tank **58**. For example, the first outer side **90** of the skirt **74** includes threads **110** configured to cooperate with threads **112** on an inner side of a neck **113** of the hydraulic fluid tank **58** (see FIG. **4**) to releasably couple the breather cap **66** to the hydraulic fluid tank **58**. In some embodiments, the breather cap **66** also includes one or more torquing features on

the base **72** of the breather cap **66** configured to be engaged with a tool to tighten or engage or loosen/disengage the breather cap **66** on the hydraulic fluid tank **58**. For example, as shown in FIGS. **5** and **7**, the torquing features include four blind holes **114** extending from the first side **78** of the base **72** toward the second side **80** of the base **72**. Although four holes **114** are shown, it should be understood that greater or fewer than four holes **114** may be included without departing from the scope of the disclosure. The four holes **114** are configured to be engaged with a special tool that is not readily available to end users, such as owners of the vehicle **10**, but may be available to vehicle assemblers, dealers, and/or mechanics.

[0060] With reference to FIGS. **7** and **8**, the exemplary breather cap **66** includes a plurality of mounting grooves **116** located on the second inner side **92** of the skirt **74**. For example, the breather cap **66** includes four mounting grooves **116**. In some embodiments, each mounting groove **116** includes an axial groove **118** and a circumferential groove **120**, which are connected and cooperate to form an L-shaped mounting groove **116**. For example, the axial groove **118** is on the second inner side **92** of the skirt **74** and extends from the distal end **88** of the skirt **74** toward the second side **80** of the base **72**. The axial groove **118** terminates at or proximate to the second shoulder **102**. The axial groove **118** has a first width **122**. The circumferential groove **120** extends at least partially circumferentially along the second inner side **92** of the skirt **74** from the axial groove **118** and terminates in a groove stop **124**. The circumferential groove **120** has a second width **126**.

[0061] In some embodiments, one, some, or all of the mounting grooves **116** include an optional detent **128** that extends axially away from the second shoulder **102** proximate to or at the groove stop **124**. The detent **128** has a third width **130** that is slightly greater than the second width **126** of the circumferential groove **120**. In some embodiments, the circumferential groove **120** includes a lead in ramp **132** proximate to the connection of the axial groove **118** and the circumferential groove **120**. The circumferential groove **120** extends along the second shoulder **102** and is bounded by the second shoulder **102** on one side.

[0062] Stated another way, and as shown in FIGS. **6** and **8**, each mounting groove **116** is formed by the second shoulder **102** of the skirt **74** and an L-shaped rib **134** on the second inner side **92** of the skirt **74**. The L-shaped rib **134** includes an axial rib **136** and a circumferential rib **138** connected thereto. The axial rib **136** extends from the second shoulder **102** toward the distal end **88** of the skirt **74**. The axial rib **136** has a first axial wall **140** and a second axial wall **142**. The circumferential rib **138** extends at least partially circumferentially along the second inner side **92** of the skirt **74** from the axial rib **136** away from the first axial wall **140** of the axial rib **136** and terminates in a third axial wall **144**. The circumferential rib **138** has a circumferential wall **146** located between the second shoulder **102** and the distal end **88** of the skirt **74**. In some embodiments, the circumferential rib **138** has a recessed portion **148** proximate to or at the intersection of the axial rib **136** and the circumferential rib **138**, the recessed portion **148** located circumferentially between the second axial wall **142** and the circumferential wall **146**. The circumferential rib **138** may include a chamfer **150** proximate to the between the circumferential wall **146** and the third axial wall **144**, proximate to the third axial wall **144**. Opposite the circumferential wall **146**, the circumferential rib **138** may have a taper **152** or may taper toward the distal end **88** of the skirt **74**.

[0063] The circumferential wall **146** of the circumferential rib **138** and the second shoulder **102** cooperate to form the circumferential groove **120**. The first axial wall **140** of the axial rib **136** and the third axial wall **144** of an adjacent circumferential rib **138** cooperate to form the axial groove **118**. The second axial wall **142** of the axial rib **136** forms the groove stop **124** of the circumferential groove **120**. The recessed portion **148** of the circumferential rib **138** forms the detent **128** of the mounting groove **116**. The distance between the circumferential wall **146** and the second shoulder **102** forms the second width **126** of the circumferential groove **120** and the distance between the recessed portion **148** and the second shoulder **102** forms the third width **130** of the detent **128**. The chamfer **150** in the circumferential rib **138** forms the lead in ramp **132** of the

mounting groove **116**.

Filter

[0064] With continued reference to FIGS. **5**, **6**, and **7**, an exemplary filter **68** is illustrated. The filter **68** includes a filter element **162** and a filter edge finishing **160** coupled to the filter element **162**. The filter element **162** may have one or more openings **170** through which gas (e.g., air) is able to pass, and into and out of the chamber **64** of the hydraulic fluid tank **58**, while filtering out contaminants. Thus, the filter **68** is configured to prevent contaminants from entering hydraulic fluid tank **58** that could compromise or reduce the performance of the hydraulic fluid and the suspension system **12**. In some embodiments, for example only and without limitation, the filter element **162** may be a filtration grade 40 μm 3-layer sintered wire mesh laminate, wherein the wire mesh is AISI 304/316, and has an air permeation capability of greater than 1.5 L/min.

[0065] The filter edge finishing **160** has a first side **164**, a second side **166** opposite the first side **164**, and a thickness **168** between the first side **164** and the second side **166**. In some embodiments, the filter edge finishing **160** and the filter element **162** are circular in shape, with the filter edge finishing **160** having an annular or ring shape extending circumferentially around the filter element **162**. Although the filter element **162** and filter edge finishing **160** are shown as circular, the filter element **162** and/or the filter edge finishing **160** may have other shapes without departing from the scope of the disclosure. In some embodiments, the filter edge finishing **160** is a metal ring that is folded and crimped around the edge of the filter element **162**. In some embodiments, the filter edge finishing **160** may be a resilient or compressible material, such as for example, rubber, that may be molded onto the filter element **162**. In some embodiments, the filter edge finishing **160** may be a gasket. The filter **68** is sized to be disposed in the chamber **98** of the skirt **74** on the second inner side **92** of the skirt **74** and is configured to rest against the second shoulder **102** of the skirt **74**.

Splash Guard

[0066] With reference to FIGS. **5**, **6**, **7**, and **9**, an exemplary splash guard **70** is illustrated. The splash guard **70** includes a base **174** and a guard portion **176** extending from the base **174**, and a distal opening **178** extending through the guard portion **176**. The guard portion **176** is raised with respect to the base **174**. The guard portion **176** is configured to extend toward the chamber **64** of the hydraulic fluid tank **58** (see FIG. **13**). The base **174** has a first side **180**, a second side **182** opposite the first side **180**, and a thickness **184** between the first side **180** and the second side **182**. In some embodiments, the base **174** is circular in shape and includes one or more tabs **186** extending radially outward from the base **174**. The base **174** is sized to be disposed in the chamber **98** of the skirt **74** on the second inner side **92** of the skirt **74** and is configured to rest against the filter **68**. The one or more tabs **186** are configured to be received in the one or more mounting grooves **116** in the breather cap **66** to releasably couple the splash guard **70** with the breather cap **66**. For example, the tabs **186** may have a width **187**, wherein the width is less than or equal to the first width **122** of the axial grooves **118** of the breather cap **66**. The exemplary splash guard **70** includes four tabs **186**, with each tab **186** spaced equally circumferentially around the base **174**, such as for example, **90** degrees apart. However it will be understood that greater or fewer numbers of tabs **186** may be included without departing from the scope of the disclosure. Additionally, in some embodiments, the tabs **186** may not be spaced equally circumferentially around the base **174** without departing from the scope of the disclosure.

[0067] The guard portion **176** of the splash guard **70** extends a first distance **188** from the second side **182** of the base **174** and away from the first side **180** of the base **174**. The guard portion **176** terminates in a distal end **190**. The guard portion **176** is wider at the base **174** than at the distal end **190**. The distal opening **178** is located at the distal end **190** of the guard portion **176**. The guard portion **176** includes a first outer side **192** and a second inner side **194**. Proximate to the first side **180** of the base **174** is a proximal opening **196** to an open chamber **198** formed by the second inner side **194** of the guard portion **176**. The distal opening **178** has a diameter **197** less than the diameter **199** of the proximal opening **196**.

[0068] As shown in FIGS. 5, 6, and 7, the guard portion **176** has a domed shape. In some embodiments, the guard portion **176** may have a hemispherical shape. However, it will be understood that in other embodiments, the guard portion **176** may have other shapes without departing from the scope of the disclosure, for example and without limitation, a conical shape, a frustoconical shape, a cylindrical shape, and an hourglass shape.

[0069] In some embodiments, the splash guard **70** further includes one or more features configured to be engaged with a tool to engage or disengage the splash guard **70** with the breather cap **66**. As shown in FIGS. 6, 7, and 9, the splash guard **70** includes one or more lugs **200** extending from the second side **182** of the base **174**. In some embodiments, the lugs **200** are integrally formed with the base **174** and the guard portion **176** of the splash guard **70**. As shown in FIG. 9, the lugs **200** may form a discontinuous square shape around the guard portion **176**. In some embodiments, the lugs **200** are configured to be engaged with a special tool that is not readily available to end users, such as owners of the vehicle **10**, but may be available to vehicle assemblers, dealers, and/or mechanics. In some embodiments, the lugs **200** may be configured to be engaged with by a standard tool.

Assembly of the Breather Cap Assembly

[0070] Having described the individual components of the breather cap assembly **60**, the assembly thereof is described. To assemble the breather cap assembly **60**, the filter **68** is inserted into the skirt **74** of the breather cap **66** through the distal opening **97** and is pushed toward the second side **80** of the breather cap **66** until the filter edge finishing **160** contacts the second shoulder **102**. The filter **68** is thus seated against the second shoulder **102** of the breather cap **66**. Then the splash guard **70** is oriented with the first side **180** of the base **174** facing the second side **80** of the breather cap **66**, such that the guard portion **176** is facing away from the second side **80** of the breather cap **66**. The tabs **186** of the splash guard **70** are then oriented to align with the axial grooves **118** of the breather cap **66**. The tabs **186** of the splash guard **70** are then inserted into the axial grooves **118** of the breather cap **66**, with one tab **186** per axial groove **118**. The splash guard **70** is then pushed toward the second side of the filter **68** (along arrow A in FIG. 11) until the first side **180** of the splash guard **70** contacts the filter edge finishing **160**. The splash guard **70** is then rotated (along arrow C in FIG. 11) such that the tabs **186** of the splash guard **70** move along and within the circumferential grooves **120** of the breather cap **66**. The lead in ramp **132** assists with the transition of the movement of the tabs **186** of the splash guard **70** from axial movement (along arrow A) in the axial grooves **118** to circumferential movement (along arrow C) in the circumferential grooves **120**. The splash guard **70** is rotated until the tabs **186** contact the groove stops **124** of the circumferential grooves **120**, which are configured to prevent further rotation of the splash guard **70**. In some embodiments, the filter edge finishing **160** is compressed against the second shoulder **102** by the splash guard **70** and a sealing contact may be formed between the filter edge finishing **160**, the second shoulder **102** and the splash guard **70**. The splash guard **70** retains the filter **68** in the open chamber or recess **98** of the breather cap **66**. In the event that disassembly of the breather cap assembly **60** is desired or required, for example to service, clean, and/or replace one or more of the breather cap **66**, filter **68**, and splash guard **70**, the breather cap assembly **60** may be disassembled by rotating the splash guard **70** in the opposite direction of arrow C, removing the splash guard **70** from breather cap **66** along the opposite direction of arrow A. The filter **68** may then be removed from the breather cap **66**. For example, when the filter **68** has reached the end of its service life, it may be removed and replaced with a new filter **68**.

[0071] As described above, in some embodiments, the breather cap **66** may optionally include detents **128** into which the tabs **186** of the splash guard **70** are configured to interface with. In some embodiments, the combined thickness of the filter edge finishing **160** and the base **174** of the splash guard **70** (i.e., the sum of thickness **168** of the filter edge finishing **160** and the thickness **184** of the splash guard **70**) may be slightly greater than the third width **130**. During assembly of the splash guard **70** into the breather cap **66**, the filter edge finishing **160** may be slightly compressed. When the tabs **186** of the splash guard **70** reach and enter into the detent **128**, the filter edge

finishing **160** is able to expand pressing the tabs **186** of the splash guard **70** into the detents **128**, assisting in retaining the tabs **186** in the detents **128** and the splash guard **70** in the breather cap **66**. At that point, the filter **68** is sandwiched between the breather cap **66** and the splash guard **70**, with the first side **164** of the filter edge finishing **160** contacting the second shoulder **102** of the skirt **74** and the second side **166** of the filter edge finishing **160** contacting the first side **180** of the base **174** of the splash guard **70**, this contact forms a seal between the breather cap **66**, the filter edge finishing **160**, and the splash guard **70**.

[0072] As shown in FIG. 4, the breather cap assembly **60** is releasably coupled to the hydraulic fluid tank **58** by the threads **110** of the breather cap **66** and the threads **112** of the neck **113** of the hydraulic fluid tank **58**. When the breather cap **60** is coupled to the hydraulic fluid tank **58**, the gasket **62** is sandwiched between the second side **80** of the base **72** of the breather cap **66** and the neck **113** of the hydraulic fluid tank **58** to form a seal between the breather cap assembly **60** and the hydraulic fluid tank **58**.

[0073] In some embodiments, at least a portion of the splash guard **70** extends beyond the breather cap **66** toward the chamber **64** of the hydraulic fluid tank **58**. For example, as shown in FIG. 12, when the splash guard **70** is installed in the breather cap **66**, in some embodiments, the distal end **190** of splash guard **70** extends a distance **202** beyond distal end **88** of breather cap **66**. However, it should be understood that in some embodiments, the distal end **190** of splash guard **70** does not extend beyond the distal end **88** of breather cap **66**. In some embodiments, the distance **202** is zero. In some embodiments, the distal end **88** of breather cap **66** extends a distance beyond the distal end **190** of splash guard **70**.

[0074] As shown in FIGS. 4 and 12, the breather cap **66**, filter **68**, and filter edge finishing **160** cooperate to form a breather cap assembly **60** through which gas (e.g., air) is able to pass, and into and out of the chamber **64** of the hydraulic fluid tank **58** as shown by bi-directional arrow G. For example, gas (e.g., air) may pass through the breather cap assembly **60** via passageway **76** and open chamber **98** of the breather cap **66**, openings **170** in the filter **68**, and open chamber **198** and distal opening **178** of splash guard **70**.

Alternative Embodiment of Breather Cap Assembly with Dipstick

[0075] Now with continued reference to FIG. 12, the breather cap assembly **60** is shown in which the splash guard **70** includes an optional dipstick **204**. The dipstick **204** is coupled to the splash guard **70** and extends away from the base **174** of the splash guard **70**. For example, the dipstick **204** is coupled to the guard portion **176** of the splash guard **70**. The dipstick **204** includes markings **206** proximate to a distal end **208** of the dipstick **204** that are configured to assist in determining the level of hydraulic fluid contained in the hydraulic fluid tank **58**. In some embodiments, the dipstick **204** is integrally formed with the splash guard **70**.

Alternative Embodiment of the Breather Cap Assembly

[0076] Now with reference to FIGS. 13-18, an alternative embodiment of the breather cap assembly **60** is shown. The breather cap assembly **60** includes a breather cap **66**, a filter **68**, and a splash guard **70**. The filter **68** is sandwiched between the breather cap **66** and the splash guard **70**.

Alternative Embodiment of the Breather Cap

[0077] The breather cap **66** includes a base **72**, a first skirt **74** extending from the base **72**, and a passageway **76** extending through the base **72**. The base **72** has a first side **78**, a second side **80** opposite the first side **78**, and a thickness **82** between the first side **78** and the second side **80**. In some embodiments, the base **72** is cylindrical. Although the base **72** is shown as cylindrical, the base **72** may have other shapes without departing from the scope of the disclosure.

[0078] The first skirt **74** extends a first distance **86** from the second side **80** of the base **72** and away from the first side **78** of the base **72**. The first skirt **74** terminates in a distal end **88**. The exemplary first skirt **74** is cylindrical and includes a first outer side **90** and a second inner side **92**.

[0079] At the distal end **88** of the first skirt **74** is a distal opening **97** to an open chamber **98** formed by the first skirt **74**. Additionally, at the distal end **88** of the first skirt **74** is an annular end surface

100 extending between the first outer side **90** and the second inner side **92**.

[0080] The breather cap **66** further includes a second skirt **210** within the open chamber **98** formed by the first skirt **74** and extending from the base **72** a second distance **212** from the second side **80** of the base **72** and away from the first side **78** of the base **72**. The first skirt **74** may be referred to as an outer skirt and the second skirt **210** may be referred to as an inner skirt. The second skirt **210** terminates in a distal end **214**. The exemplary second skirt **210** is cylindrical and includes a first outer side **216** and a second inner side **218**. The second skirt **210** is concentric with the first skirt **74**. At the distal end **214** of the second skirt **210** is a distal opening **220** to an open chamber **222** formed by the second skirt **210**. Additionally, at the distal end **214** of the second skirt **210** is an annular end surface **224** extending between the first outer side **216** and the second inner side **218** of the second skirt **210**. In some embodiments, a circumferential chamfer **225** is located on the second skirt **210** between the first outer side **216** and the annular end surface **224**. The second distance **212** is less than the first distance **86**. Thus, in some embodiments, the first skirt **74** extends further from the base **72** than the second skirt **210**. Stated another way, the second skirt **210** is shorter than the first skirt **74**.

[0081] The first outer side **216** of the second skirt **210** faces the second inner side **92** of the first skirt **74**. The second skirt **210** includes a groove **226** extending at least partially circumferentially around the first outer side **216** of the second skirt **210** proximate to the second side **80** of the base **72** of the breather cap **66**. In some embodiments, the groove **226** extends entirely around the circumference of the second skirt **210**. In some embodiments, the groove **226** extends only partially around the circumference of the second skirt **210**. In some embodiments, the groove **226** is discontinuous in its extension around circumference of the second skirt. In such embodiments, the groove **226** may be said to include a plurality of discontinuous grooves **226** extending at least partially around the circumference of the second skirt **210**.

[0082] The breather cap **66** includes threads **110** configured to cooperate with the threads **112** on the hydraulic fluid tank **58**. The threads **110** are located on the second inner side **92** of the first skirt **74**. As shown in FIG. **13**, the threads **112** are located on an outer side of the neck **113** of the hydraulic fluid tank **58**.

[0083] The breather cap **66** also includes one or more one or more torquing features on the breather cap **66** configured to be engaged with a tool to tighten or engage or loosen/disengage the breather cap **66** with the hydraulic fluid tank **58**. For example, as shown in FIGS. **14**, **15**, **16**, and **18**, the torquing features include four flat portions **228** formed on the first skirt **74** proximate to the first side **78** of the base **72**. Although four flat portions **228** are shown, it should be understood that greater or fewer than four flat portions **228** may be included without departing from the scope of the disclosure. In some embodiments, the four flat portions **228** are configured to be engaged with a special tool that is not readily available to end users, such as owners of the vehicle **10**, but may be available to vehicle assemblers, dealers, and/or mechanics. In other embodiments, the four flat portions **228** may be configured to be engaged with a generally-available wrench.

Alternative Embodiment of the Splash Guard

[0084] With continued reference to FIGS. **13-17**, another exemplary splash guard **70** is illustrated. The splash guard **70** includes a base **174**, a guard portion **176** extending from the base **174**, and a distal opening **178** extending through the guard portion **176**. The guard portion **176** is raised with respect to the base **174**. The guard portion **176** is configured to extend toward the chamber **64** of the hydraulic fluid tank **58** (see FIG. **4**). The base **174** has a first side **180**, a second side **182** opposite the first side **180**, and a thickness **184** between the first side **180** and the second side **182**. In some embodiments, the base **174** is circular in shape. The embodiment of the splash guard **70** shown in FIGS. **13-17** includes one or more fingers **240** extending from the first side **180** of the base **174** away from the guard portion **176**. The fingers **240** terminate in a distal end **242**. The fingers **240** are curved and cooperate to form a cylindrical skirt shape. At the distal end **242** of each finger **240** is a protrusion **244** that extends radially inward from the finger **240**. The protrusions **244**

are configured to be received in the groove **226** of the breather cap **66** to retain the splash guard **70** on the breather cap **66**.

[0085] The fingers **240** are configured to bent outward slightly during assembly of the splash guard **70** with the breather cap **66** and are then configured to return to their un-bent state when the protrusions **244** are received in the groove **226** of the breather cap **66**. The protrusions **244** are configured to slide up the circumferential chamfer **225** on the second skirt **210** of the breather cap **66**, to aid in assembly of the splash guard **70** on the breather cap **66**.

[0086] The fingers **240** are at least partially separated by slots **246** that extend from the distal end **242** of the fingers toward the first side **180** of the base **174**. As shown in FIGS. **14**, **15**, and **16**, the splash guard **70** includes a web **248** between each pair of fingers **240**. In some embodiments, the slots **246** extend all the way to the first side **180** of the base **174** and therefore lack the webs **248**. Although four fingers **240** are shown, it should be understood that greater or fewer than four fingers **240** may be included without departing from the scope of the disclosure. At the distal end **242** of the fingers **240** is a distal opening **250** to an open chamber **252** formed by the fingers **240**. The fingers **240** are located on the splash guard **70** to permit the second skirt **210** of the breather cap **66** to be received in the open chamber **252** formed by the fingers **240**. The fingers **240** of the splash guard **70** are configured to be received in the annular space between the first skirt **74** and the second skirt **210** of the breather cap **66**.

[0087] The guard portion **176** of the splash guard **70** extends the first distance **188** from the second side **182** of the base **174** and away from the first side **180** of the base **174**. The guard portion **176** terminates in the distal end **190**. The distal opening **178** is located at the distal end **190** of the guard portion **176**. The guard portion **176** includes the first outer side **192** and the second inner side **194**. Proximate to the first side **180** of the base **174** is the proximal opening **196** to the open chamber **198** formed by the second inner side **194** of the guard portion **176**. The distal opening **178** has a diameter **197** less than the diameter **199** of the proximal opening **196**. The open chamber **198** of the guard portion **176** and the open chamber **252** of the fingers **240** are fluidly connected. Accordingly, a fluid may pass through distal opening **250**, through open chamber **252**, through proximal opening **196**, through open chamber **198**, and through distal opening **178**.

[0088] The guard portion **176** has a conical shape. In some embodiments, the guard portion **176** may have a hemispherical shape. However, it will be understood that in other embodiments, the guard portion **176** may have other shapes without departing from the scope of the disclosure.

[0089] The filter **68** is sized to be disposed in the open chamber **252** of the fingers **240** on the first side **180** of the base **174** of the splash guard **70** and is configured to rest against the first side **180** of the base **174** of the splash guard **70**.

Assembly of the Alternative Embodiment of the Breather Cap Assembly

[0090] Having described the individual components of the breather cap assembly **60** shown in FIGS. **13-18**, the assembly thereof is described. To assemble the breather cap assembly **60**, the filter **68** is inserted into the open chamber **252** of the fingers **240** on the first side **180** of the base **174** of the splash guard **70**. The second side **166** of the filter edge finishing **160** contacts the first side **180** of the base **174**. Then the splash guard **70** is oriented with the first side **180** of the base **174** facing the second side of the breather cap **66**, such that the guard portion **176** is facing away from the second side of the breather cap **66**.

[0091] The breather cap **66** and the splash guard **70** are then moved closer to one another such that the fingers **240** are brought toward the second side **80** of the base **72** of the breather cap **66**. After the distal ends **242** of the fingers **240** reach the distal end **214** of the second skirt **210**, the protrusions **244** contact the circumferential chamfer **225** and then the first outer side **216** of the second skirt **210**, causing the fingers **240** to bend radially outward slightly. When the protrusions **244** reach the groove **226** in the second skirt **210**, the protrusions **244** seat into the groove **226** and the fingers **240** return to their un-bent state. At that point, the filter **68** is sandwiched between the breather cap **66** and the splash guard **70**, with the first side **164** of the filter edge finishing **160**

contacting the distal end **214** of the second skirt **210** and the second side **166** of the filter edge finishing **160** contacting the first side **180** of the base **174** of the splash guard **70**, this contact forms a seal between the breather cap **66**, the filter edge finishing **160**, and the splash guard **70**. The filter **68** is thus seated against the annular end surface **224** of the second skirt **210** of the breather cap **66**. [0092] In some embodiments, at least a portion of the splash guard **70** extends beyond the breather cap **66** toward the chamber **64** of the hydraulic fluid tank **58**. For example, as shown in FIG. **17**, when the splash guard **70** is installed in the breather cap **66**, in some embodiments, the distal end **190** of splash guard **70** extends a distance **202** beyond distal end **88** of breather cap **66**. However, it should be understood that in some embodiments, the distal end **190** of splash guard **70** does not extend beyond the distal end **88** of breather cap **66**. In some embodiments, the distance **202** is zero. In some embodiments, the distal end **88** of breather cap **66** extends a distance beyond the distal end **190** of splash guard **70**.

[0093] As shown in FIG. **13**, the breather cap assembly **60** is releasably coupled to the hydraulic fluid tank **58** by the threads **110** of the breather cap **66** and the threads **112** of the neck **113** of the hydraulic fluid tank **58**. When the breather cap **60** is coupled to the hydraulic fluid tank **58**, the splash guard **70** is sandwiched between the neck **113** of the hydraulic fluid tank **58** and the breather cap **66**. For example, the second side **182** of the base **174** of the splash guard **70** is in contact with an annular end surface **260** of the neck **113**. Thus, the breather cap **66** urges the base **174** of the splash guard **70** toward the neck **113**. Additionally, the neck **113** of the hydraulic fluid tank **58** may include a circumferential groove **262** into which a gasket **62** may be disposed. When the breather cap **60** is coupled to the hydraulic fluid tank **58**, the second inner side **92** of the skirt **74** and the gasket **62** contact one another to form a seal between the breather cap assembly **60** and the hydraulic fluid tank **58**. Additionally, in some embodiments, when the breather cap **60** is coupled to the hydraulic fluid tank **58**, the second side **182** of the base **174** of the breather cap **66** and annular end surface **260** of the neck **113** of the hydraulic fluid tank **58** may cooperate to form a seal between the breather cap assembly **60** and the hydraulic fluid tank **58**.

[0094] Additionally, as shown in FIGS. **13** and **17**, the breather cap **66**, filter **68**, and filter edge finishing **160** cooperate to form a breather cap assembly **60** through which gas (e.g., air) is able to pass, and into and out of the chamber **64** of the hydraulic fluid tank **58** as shown by bi-directional arrow **G**. For example, gas (e.g., air) may pass through the breather cap assembly **60** via passageway **76** and open chamber **222** of the breather cap **66**, openings **170** in the filter **68**, and open chamber **198** and distal opening **178** of splash guard **70**.

Operation of Breather Cap Assemblies

[0095] During operation of the vehicle **10** in extreme driving conditions, for example up and/or down steep inclines, off-road, or on racetracks, the hydraulic fluid sloshes around in the chamber **64** of the hydraulic fluid tank **58**. When the breather cap assembly **60** is assembled with the hydraulic fluid tank **58**, the splash guard **70** is located between the filter **68** and the hydraulic fluid in the hydraulic fluid tank **58** and the guard portion **176** of the splash guard **70** extends toward the chamber **64** of the hydraulic fluid tank **58**. The guard portion **176** of the splash guard **70** reduces or prevents the potential for hydraulic fluid to splash and contact the filter **68**. The diameter of the distal opening **178** is large enough for gas to pass through, but small enough to reduce or eliminate the potential for hydraulic fluid to pass through the distal opening **178** and contact the filter **68**. Additionally, because the guard portion **176** is wider at the base **174** than at the distal end **190** and because the guard portion **176** is oriented with the distal end **190** between the hydraulic fluid and the base **174**, if any hydraulic fluid were to pass through the distal opening **178** and enter into the open chamber **198** of the splash guard **70**, the hydraulic fluid could be drawn by gravity down the second inner side **194** of the splash guard **70** toward the distal opening **178** and drain out of the distal opening **178**.

[0096] The foregoing description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. It is not intended to be exhaustive or to limit the invention.

Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the invention, and all such modifications are intended to be included within the scope of the invention. Thus, the broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims. Thus, many other modifications and variations of the present disclosure are possible in light of the above teachings and may be practiced otherwise than as specifically described while within the scope of the appended claims. It should be understood that one or more steps within a method may be executed in different order (or concurrently) without altering the principles of the present disclosure. Further, although each of the embodiments is described above as having certain features, any one or more of those features described with respect to any embodiment of the disclosure can be implemented in and/or combined with features of any of the other embodiments, even if that combination is not explicitly described. In other words, the described embodiments are not mutually exclusive, and permutations of one or more embodiments with one another remain within the scope of this disclosure.

Claims

1. A hydraulic fluid tank assembly for use with a suspension system for a vehicle, the hydraulic fluid tank assembly comprising: a tank, the tank having a chamber configured to contain a volume of hydraulic fluid and a volume of gas; and a breather cap assembly releasably coupled to the tank, the breather cap assembly comprising: a breather cap, the breather cap having a passageway and a recess; a filter, the filter having one or more openings configured to permit gas to pass therethrough; and a splash guard coupled to the breather cap, the filter disposed between the splash guard and the breather cap, the splash guard retaining the filter in the recess of the breather cap, and the splash guard having a distal opening, the passageway of the breather cap, the openings of the filter, and the distal opening of the splash guard fluidly connected and configured to permit gas to pass therethrough to and from the chamber of the tank and the atmosphere.
2. The hydraulic fluid tank assembly of claim 1, the breather cap comprising: a base having a first side and a second side opposite the first side; a cylindrical skirt extending from the second side of the base and away from the first side, the cylindrical skirt terminating at a distal end, the cylindrical skirt having an inner side forming the recess, and a shoulder on the inner side of the cylindrical skirt, the shoulder located a distance from the second side of the base between the second side of the base and the distal end of the cylindrical skirt, the filter being seated against the shoulder of the breather cap; and the passageway extending through the base from the first side of the base to the second side of the base.
3. The hydraulic fluid tank assembly of claim 2, the breather cap further comprising threads on an outer side of the cylindrical skirt.
4. The hydraulic fluid tank assembly of claim 1, the breather cap further comprising a nipple extending from the breather cap, the passageway extending through the nipple.
5. The hydraulic fluid tank assembly of claim 1, the breather cap further comprising one or more features on the breather cap configured to be engaged with by a tool to engage or disengage the breather cap with the tank.
6. The hydraulic fluid tank assembly of claim 5, the one or more features comprising four blind holes.
7. The hydraulic fluid tank assembly of claim 1, the splash guard comprising: a base having a first side and a second side opposite the first side; and a guard portion extending from the second side of

the base away from the first side, the distal opening extending through the guard portion.

8. The hydraulic fluid tank assembly of claim 7, wherein the base is circular and the splash guard includes one or more tabs extending radially outward from the base, the one or more tabs configured to be releasably engaged with the breather cap to releasably couple the splash guard with the breather cap.

9. The hydraulic fluid tank assembly of claim 7, the splash guard further comprising one or more features configured to be engaged with by a tool to engage or disengage the splash guard with the breather cap.

10. The hydraulic fluid tank assembly of claim 1, wherein at least a portion of the splash guard extends beyond the breather cap toward the chamber of the tank.

11. The hydraulic fluid tank assembly of claim 2, the breather cap further comprising a mounting groove, the mounting groove comprising: an axial groove on the inner side of the cylindrical skirt, the axial groove extending from the distal end of the cylindrical skirt toward the second side of the base, the axial groove terminating proximate to the shoulder; and a circumferential groove on the inner side of the cylindrical skirt, the circumferential groove extending at least partially circumferentially around the inner side of the cylindrical skirt, the circumferential groove connected to the axial groove.

12. The hydraulic fluid tank assembly of claim 11, the mounting groove further comprising a detent connected to the circumferential groove.

13. A hydraulic fluid tank assembly for use with a suspension system for a vehicle, the hydraulic fluid tank assembly comprising: a tank, the tank having a chamber configured to contain a volume of hydraulic fluid and a volume of gas; and a breather cap assembly releasably coupled to the tank, the breather cap assembly comprising: a breather cap, the breather cap comprising: a base having a first side and a second side opposite the first side, an outer cylindrical skirt extending a first distance from the second side of the base away from the first side, an inner side of the outer cylindrical skirt having a retaining feature adapted to releasably couple the breather cap to the tank; an inner cylindrical skirt extending a second distance from the second side of the base away from the first side, the second distance less than the first distance; and a passageway extending through the base; a filter, the filter having one or more openings configured to permit gas to pass therethrough; and a splash guard having a distal opening, the splash guard coupled to the inner cylindrical skirt of the breather cap, with the filter disposed between the splash guard and the breather cap; the passageway of the breather cap, the openings of the filter, and the distal opening of the splash guard fluidly connected and configured to permit gas to pass therethrough to and from the chamber of the tank and the atmosphere.

14. The hydraulic fluid tank assembly of claim 13, wherein the breather cap has a groove on an outer side of the inner cylindrical skirt, and wherein the splash guard further comprises: a base having a first side and a second side opposite the first side; a guard portion extending from the second side of the base away from the first side; and one or more fingers extending from the first side of the base away from the second side, each finger having a distal end and a protrusion proximate to the distal end, each protrusion seated in the groove of the breather cap.

15. The hydraulic fluid tank assembly of claim 14, wherein the one or more fingers are curved.

16. The hydraulic fluid tank assembly of claim 14, wherein the guard portion has a conical shape.

17. The hydraulic fluid tank assembly of claim 14, the inner cylindrical skirt having a distal end and an annular surface at the distal end, the filter being sandwiched between the annular surface and the first side of the base of the splash guard.

18. The hydraulic fluid tank assembly of claim 13, wherein the retaining feature comprises threads configured to interface with threads on the tank.

19. The hydraulic fluid tank assembly of claim 13, the breather cap further comprising one or more features on the breather cap configured to be engaged with by a tool to engage or disengage the breather cap with the tank.

20. A hydraulic fluid tank assembly for use with a suspension system for a vehicle, the hydraulic fluid tank assembly comprising: a tank, the tank having a chamber configured to contain a volume of hydraulic fluid and a volume of gas; and a breather cap assembly releasably coupled to the tank, the breather cap assembly comprising: a breather cap, the breather cap comprising: a base having a first side and a second side opposite the first side; a cylindrical skirt extending from the second side of the base and away from the first side, the cylindrical skirt having a first outer side, a second inner side, and a shoulder on the second inner side of the cylindrical skirt, the shoulder located a distance from the second side of the base; one or more grooves on the second inner side of the cylindrical skirt, the one or more grooves proximate to the shoulder; and a passageway extending through the base from the first side of the base to the second side of the base; a filter, the filter having a filter edge finishing and a filter element, the filter edge finishing in contact with the shoulder of the breather cap, the filter configured to permit gas to pass therethrough; and a splash guard coupled to the breather cap, the filter disposed between the splash guard and the breather cap, the splash guard comprising: a circular base having a first side and a second side opposite the first side; one or more tabs extending radially from circular base, the one or more tabs located in the one or more grooves of the breather cap; a guard portion extending from the second side of the base away from the first side, the guard portion extending toward the chamber of the tank; and a distal opening extending through the guard portion.
